

Chemistry LLM Evaluation Report

1. Executive Summary

- Total Evaluated Items:** 1200
- Micro Accuracy:** 94.33% (95% CI: [92.8%, 95.6%])
- Macro Accuracy (by capability):** 96.22%
- Invalid Format Rate:** 0.00%
- Numerical MAE:** 7.6552 | **RMSE:** 86.3382

2. Capability Breakdown

Capability	Count	Accuracy	95% CI
numerical_reasoning	600	88.7%	[85.8%, 91.2%]
factual_recall	500	100.0%	[100.0%, 100.0%]
stereochemical_reasoning	100	100.0%	[100.0%, 100.0%]

3. Subfield Breakdown

Subfield	Count	Accuracy	95% CI
Stoichiometry & Reactions	100	78.0%	[69.0%, 87.0%]
Electrochemistry	100	100.0%	[100.0%, 100.0%]
Atomic Structure & Trends	100	100.0%	[100.0%, 100.0%]
Organic Chemistry & Stereochemistry	100	100.0%	[100.0%, 100.0%]
Chemical Kinetics	100	82.0%	[75.0%, 89.0%]
Thermodynamics & Thermochemistry	100	100.0%	[100.0%, 100.0%]
Chemical Equilibrium	100	100.0%	[100.0%, 100.0%]
Acids, Bases & Buffers	100	100.0%	[100.0%, 100.0%]
Gas Laws & Ideal Gases	100	88.0%	[82.0%, 94.0%]
Solutions & Colligative Properties	100	84.0%	[76.0%, 91.0%]
Coordination Chemistry	100	100.0%	[100.0%, 100.0%]
Spectroscopy & Molecular Structure	100	100.0%	[100.0%, 100.0%]

4. Difficulty Breakdown

Difficulty	Count	Accuracy	95% CI
medium	700	92.6%	[90.6%, 94.4%]
easy	500	96.8%	[95.2%, 98.2%]

5. Worst 20 Failures Sample

ID	Question	Gold Answer	Predicted	Reason
sft_stoich_0003	Calculate the moles of CO ₂ produced by complete co...	9	10	numeric_mismatch (diff=1)
sft_stoich_0008	Calculate the moles of CO ₂ produced by complete co...	24	29	numeric_mismatch (diff=5)
sft_stoich_0015	Calculate the moles of CO ₂ produced by complete co...	45	35	numeric_mismatch (diff=10)
sft_stoich_0017	Calculate the moles of CO ₂ produced by complete co...	51	41	numeric_mismatch (diff=10)
sft_stoich_0024	Calculate the moles of CO ₂ produced by complete co...	72	62	numeric_mismatch (diff=10)
sft_stoich_0027	Calculate the moles of CO ₂ produced by complete co...	81	61	numeric_mismatch (diff=20)
sft_stoich_0028	Calculate the moles of CO ₂ produced by complete co...	84	76	numeric_mismatch (diff=8)
sft_stoich_0029	Calculate the moles of CO ₂ produced by complete co...	87	67	numeric_mismatch (diff=20)
sft_stoich_0030	Calculate the moles of CO ₂ produced by complete co...	90	60	numeric_mismatch (diff=30)
sft_stoich_0031	Calculate the moles of CO ₂ produced by complete co...	93	63	numeric_mismatch (diff=30)
sft_stoich_0032	Calculate the moles of CO ₂ produced by complete co...	96	66	numeric_mismatch (diff=30)
sft_stoich_0033	Calculate the moles of CO ₂ produced by complete co...	99	105	numeric_mismatch (diff=6)
sft_stoich_0044	Calculate the moles of CO ₂ produced by complete co...	132	122	numeric_mismatch (diff=10)
sft_stoich_0045	Calculate the moles of CO ₂ produced by complete co...	135	125	numeric_mismatch (diff=10)
sft_stoich_0046	Calculate the moles of CO ₂ produced by complete co...	138	128	numeric_mismatch (diff=10)
sft_stoich_0053	Calculate the moles of CO ₂ produced by complete co...	159	169	numeric_mismatch (diff=10)
sft_stoich_0080	Calculate the moles of CO ₂ produced by complete co...	240	260	numeric_mismatch (diff=20)
sft_stoich_0082	Calculate the moles of CO ₂ produced by complete co...	246	266	numeric_mismatch (diff=20)
sft_stoich_0086	Calculate the moles of CO ₂ produced by complete co...	258	298	numeric_mismatch (diff=40)
sft_stoich_0087	Calculate the moles of CO ₂ produced by complete co...	261	291	numeric_mismatch (diff=30)